Noise construction and finite-size inference in temporally correlated ballistic deposition

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Shared and independent noise innovations can change interface roughening through different driving statistics, even at the same nominal temporal-memory parameter. We isolate these effects in parallel-update ballistic deposition using stationary autoregressive controls and constructive covariance compensation. For Gaussian innovations, reweighted independent modes reproduce the shared process law, including after thresholding. For uniform innovations, matching the latent covariance leaves binary-correlation residuals of 0.039–0.052, requiring a separate match to the actual deposition field. Independent binary-covariance validation meets the retained 0.005 tolerance in 15 of 15 cells across $L = 32$–512 and $\theta = 0.10, 0.25, 0.40$. At $\theta = 0.25$, $L = 256$, a validated Gaussian surrogate still differs from the uniform-driven reference in its consecutive-four-sign moment: the difference is -0.0057 $[-0.0070, -0.0045]$. Its effective width-slope contrast across the tested sizes is -0.0175 $[-0.0365, 0.0011]$; brackets denote pointwise 95% run-bootstrap intervals. Thus a resolved high-order temporal input difference need not coincide with a resolved finite-window roughness-slope difference. The original three-input production grid and fixed-record duration controls place these contrasts in their numerical context. Our results separate innovation coupling, thresholded covariance and high-order statistics without identifying finite-size slopes with asymptotic exponents or a new universality class.

I. INTRODUCTION

The Kardar–Parisi–Zhang (KPZ) equation describes the competition between smoothing, lateral growth and stochastic driving in a fluctuating interface [1, 2],

$$\partial_t h = \nu \nabla^2 h + \frac{\lambda}{2}(\nabla h)^2 + \eta. \tag{1}$$

For short-range noise in one dimension, the roughness and dynamic exponents are $\alpha = 1/2$ and $z = 3/2$. Their stability when the noise retains temporal memory remains a problem in nonequilibrium statistical physics.

A conventional continuum description introduces the low-frequency spectrum [3, 4]

$$D(\omega, k) = D_0 + D_\theta |\omega|^{-2\theta}, \qquad 0 < \theta < 1/2, \tag{2}$$

with short-range spatial correlations and a long-time covariance contribution proportional to $|t - t'|^{2\theta-1}$. The exponent, relative amplitudes and short-time regularisation are distinct properties. A discrete fGn covariance specifies more than this long-time power. Matching its complete shape does not by itself identify a continuum universality class.

Thresholds in this literature also concern different physical questions. The nonperturbative renormalisation-group treatment finds a short-range stability boundary at $\theta = 1/6$ in one dimension [4]. Alés and López identify a boundary near 0.25 from the separation of global and spectral roughness exponents and the appearance of anomalous, faceted growth [5]. Height-distribution markers and emergence diagnostics probe other observables [6, 7]. These definitions are not interchangeable estimates of one critical parameter. Model and discretisation dependence under temporally correlated driving has also been emphasised [8].

A numerical comparison must establish which random field reaches the growth rule. An array assembly can retain correctly filtered sequences while changing whether their correlations act in space or time. A second issue arises in inference: a finite-ensemble distance has a nonzero null floor, and its first detectable deviation depends on sampling and system size. Neither issue is resolved by fitting one exponent to an entire finite-lag covariance curve.

The question addressed here is which finite-window comparisons remain interpretable when the complete input process and its implementation are specified. We validate the assembled noise field, retain the cross terms in the shared-innovation AR covariance, and compare three inputs on a matched finite-size grid. A paired analysis of the Gaussian filters uses their common random streams rather than treating their agreement as independent replication. Additional duration controls keep the spectral record fixed while extending the observation window. A separate stationary innovation campaign adds exact raw-covariance compensation and independently validated binary-covariance surrogates (Sec. IV E). Together these checks separate implementation validity, differences between specified processes, and sensitivity to observation time.

Hellinger distance and novelty detection have already been used in this setting [7], and microscopic sensitivity has been discussed [8]. Our contribution is the quantitative comparison under these explicit controls, rather than a new distance metric or filtering method. The exact nearest-neighbour increment probability mass function (pmf) also shows which morphology information a local distance retains. Our conclusions concern the tested finite windows, not an inferred critical point or a reconstruction of historical implementations.

II. MODEL AND NOISE PROCESSES

A. Parallel-update ballistic deposition

We use a periodic lattice, a flat initial condition and the parallel update

$$h(x,t+1) = \max\{h(x,t)+\eta'_\theta(x,t), h(x-1,t), h(x+1,t)\}, \tag{3}$$

where $\eta'_\theta = \mathbf{1}[\eta_\theta > 0]$ is binary deposition. This model belongs to the family studied with temporally correlated driving [5, 6, 9]. Latent sequences are independent between sites. The spectral inputs are Gaussian; the uniform-driven AR input below is not assumed to be Gaussian.

Equation (3) preserves $|h(x+1,t) - h(x,t)| \le 1$ from a flat state. Each site rises by zero or one per sweep, but can rise because of a higher neighbour even when its own deposition variable is zero. The rule does not reduce to independent deposition (Appendix A). Periodicity gives $\sum_x \Delta h = 0$, so up- and down-step counts are equal in every configuration. With $p_{\text{flat}} = P(\Delta h = 0)$, the increment pmf is

$$P(\Delta h) = \left(\frac{1-p_{\text{flat}}}{2}, p_{\text{flat}}, \frac{1-p_{\text{flat}}}{2}\right). \tag{4}$$

This is a local descriptor, not an assumed order parameter. Different spatial arrangements of the same steps can produce different large-scale interfaces.

B. Shared-innovation AR sum

The streaming construction examined here uses geometrically spaced modes,

$$u_n = aB^{-n}, \quad r_n = e^{-u_n}, \quad (a,B) = (6,2), \tag{5}$$

$$X_n(t) = r_n X_n(t-1) + z(t), \qquad \eta_\theta(t) = \sum_{n=1}^{N} W_n X_n(t), \tag{6}$$

$$W_n^2 = (1-r_n^2)K_\theta u_n^{1-2\theta}, \quad K_\theta = \frac{B^{1/2-\theta} - B^{\theta-1/2}}{\Gamma(2-2\theta)}. \tag{7}$$

Centred innovations $z(t) \sim U[-1/2, 1/2]$ are independent across time and sites but shared by all modes at a given site and time. Modes start at zero, with $N = \max[16, \lceil \log_B(aT) \rceil + 4]$. A common weight rescaling changes raw variance but not the thresholded sequence. At exactly $\theta = 0$ the simulator uses a separate i.i.d. binary reference. The unbranched AR formula remains correlated there.

We call this the *shared AR* variant. Its weights are related to the fast fractional Gaussian noise family [10], but a family name or a symbol $z_t(u_n)$ does not establish innovation covariance across modes. We do not assume every cited simulation uses these streams, initialisation or white-noise shortcut.

Writing $Q_{nm} = \langle z_n(t) z_m(t) \rangle$, the exact stationary finite-mode covariance is

$$C_N(\tau) = \sum_{n,m} W_n W_m r_n^\tau \frac{Q_{nm}}{1 - r_n r_m}, \quad \tau \ge 0. \tag{8}$$

Shared innovations have $Q_{nm} = 1/12$ for all pairs; independent innovations have $Q_{nm} = \delta_{nm}/12$, reducing Eq. (8) to a diagonal sum. These define different processes. The original production AR campaign uses the shared case only; Sec. IV E adds stationary shared, independent and covariance-compensated controls. Appendix B addresses cold starts and the interpretation of a power-law approximation.

C. Gaussian spectral filters

For each run and site we filter a length-T Gaussian white sequence in Fourier space, then threshold it. The power-law input uses

$$S_{\text{power}}(f) = |f|^{-2\theta}, \qquad f \ne 0. \tag{9}$$

The second input uses the fGn spectral shape,

$$S_{\text{fGn}}(f) \propto (1 - \cos 2\pi f) \sum_{k=-\infty}^{\infty} |2\pi(f+k)|^{-(2H+1)}, \qquad H = \theta + \tfrac{1}{2}. \tag{10}$$

The alias sum is evaluated for $k = -40, \ldots, 40$. Both spectra are sampled on a finite-record Fourier grid and have DC removed. They are finite-period approximations with a particular low-frequency cutoff, not exact infinite-fGn samples. At $\theta = 0$ the power-law spectrum is flat on nonzero modes, but DC removal still gives a finite-record anticorrelation. The truncated fGn spectrum adds a small approximation. We use *fGn-spectrum filter* with these limitations throughout.

The assembled field has shape (run, t, x): one filtered sequence occupies the complete time column of each site. A memory reshape does not exchange the physical site and time axes. New spectral archives carry an explicit corrected-layout version. Earlier unversioned spectral archives are excluded from all results below.

III. VALIDATION AND STATISTICAL PROTOCOL

A. Validation of the production field

For a centred jointly Gaussian pair with normalised covariance ρ, the centred sign field $s = 2\eta'_\theta - 1$ satisfies

$$C_s(\tau) = \langle s(t+\tau)s(t) \rangle = \frac{2}{\pi} \arcsin \rho(\tau). \tag{11}$$

The finite-period target ρ is computed by inverse Fourier transforming the actual sampled spectrum, including DC removal and alias truncation. It differs from the infinite-fGn target. Equation (11) is not imposed as an identity on the uniform-driven AR input.

The production interface is tested at $\theta = 0.25, 0.40$, $L = 32, 64$ and $T = 65536$, with eight independent runs per filter and parameter pair. Temporal products are circular to match the finite-period target; spatial products test independence at nonzero lags. Errors are standard errors across the eight run estimates. Predeclared numerical tolerances are five run standard errors plus 0.003 per reported correlation. These are engineering checks, not simultaneous confidence statements. Regression tests also cover chunk boundaries, repeated seeds and archive-version preservation.

For a negative control, the same correctly generated field is transposed to (run, x, t) and reshaped back to (run, T, L) without the proper axis mapping. This reproduces a layout fault identified during the implementation audit. It retains filtered sequences but assigns correlations to the wrong coordinates. No production result below uses this control.

B. Simulation grid and late-time width

Every input uses 64 independent runs at each combination of

$$L = 32, 64, 128, 256, 512, \tag{12}$$

$$\theta = 0, 0.10, 1/6, 0.20, 0.25, 0.30, 0.40. \tag{13}$$

There are 105 cells and 6720 runs. The AR duration is $T = \text{round}(15L^{3/2})$; spectral durations round this upwards to a 5-smooth FFT length. At $L = 32$ these are 2715 and 2880 sweeps, respectively; at $L = 512$ they are 173779 and 174960. Thus run counts, sizes and nominal duration factors are matched, with exact durations retained in each archive. Distinct runs and sites have independent streams. The two spectral filters share Gaussian base streams within a parameter cell, so their agreement is not treated as statistically independent replication.

Each run stores 32 geometric-time profiles and a regularly sampled width series $W_r^2(t) = L^{-1}\sum_x [h_r(x,t) - \bar{h}_r(t)]^2$. We average squared widths within each run over the last 20% of stored width samples and then over runs,

$$\overline{W}(L;\theta,T) = \left[\frac{1}{R}\sum_{r=1}^{R}\frac{1}{|\mathcal{T}|}\sum_{t\in\mathcal{T}} W_r^2(t)\right]^{1/2}. \tag{14}$$

The width-sampling stride is $\max(1, \lfloor T/20000 \rfloor)$ sweeps; geometric sampling applies only to profile snapshots. If n_W widths are stored, the terminal mean uses $m = \max(1, \lfloor 0.2 n_W \rfloor)$ samples, and the preceding mean uses the previous m samples. Thus these windows approximate the final and preceding fifths of elapsed time, up to stride and integer rounding. For example, the fGn-spectrum cell at $L = 256$ uses sweeps 49155–61440 for its terminal window, with a three-sweep stride. The five-size slope of $\ln \overline{W}$ against $\ln L$ is called α_{fit}. The adjacent-size slope is

$$\alpha_{\text{eff}}(L, 2L) = \frac{\ln[\overline{W}(2L)/\overline{W}(L)]}{\ln 2}. \tag{15}$$

Neither is assumed asymptotic. We compare the final 20% time window with the preceding 20% as a finite-time diagnostic rather than declaring saturation from duration alone. Widths, flat-pair fractions, window ratios and size slopes use bootstrap standard deviations from 500 resamples of independent runs. Whole run records remain together; different sizes are resampled independently. All adjacent-size estimates use the same resulting bootstrap draws. Five-point fit residuals are not substituted for run-to-run uncertainty.

C. Local distances and sampling uncertainty

We compare exact increment pmfs without rescaling their support or introducing histogram bins. The Hellinger distance is [11, 12]

$$H(P,Q) = \left[1 - \sum_j \sqrt{P_j Q_j}\right]^{1/2}. \tag{16}$$

For Eq. (4),

$$H^2 = 1 - \sqrt{p_\theta p_0} - \sqrt{(1-p_\theta)(1-p_0)}. \tag{17}$$

It contains only the flat-pair fraction and does not measure spatial arrangements or establish a universality class. For $p_\theta = p_0 + \delta p$ with $0 < p_0 < 1$, expansion gives

$$H^2 = \frac{(\delta p)^2}{8p_0(1-p_0)} + O((\delta p)^3). \tag{18}$$

This relation explains the local probability sensitivity of the distance; the distance is not an independent spatial morphology observable.

At each size and for each input, the zero-parameter ensemble is the reference. Equal halves of independent runs give the signal comparison; two disjoint reference halves give its null. The excess is

$$\Delta H = \langle H(P_{\theta,\text{half}}, P_{0,a})\rangle - \langle H(P_{0,a}, P_{0,b})\rangle, \tag{19}$$

averaged over 120 partitions. An outer bootstrap of 200 independent-run resamples repeats this complete estimator, preserving the shared reference within each resample. Its standard deviation supplies the errors. The standard deviation across conditional null partitions is a separate quantity. We define $\Delta H(0, L) = 0$ as the reference point

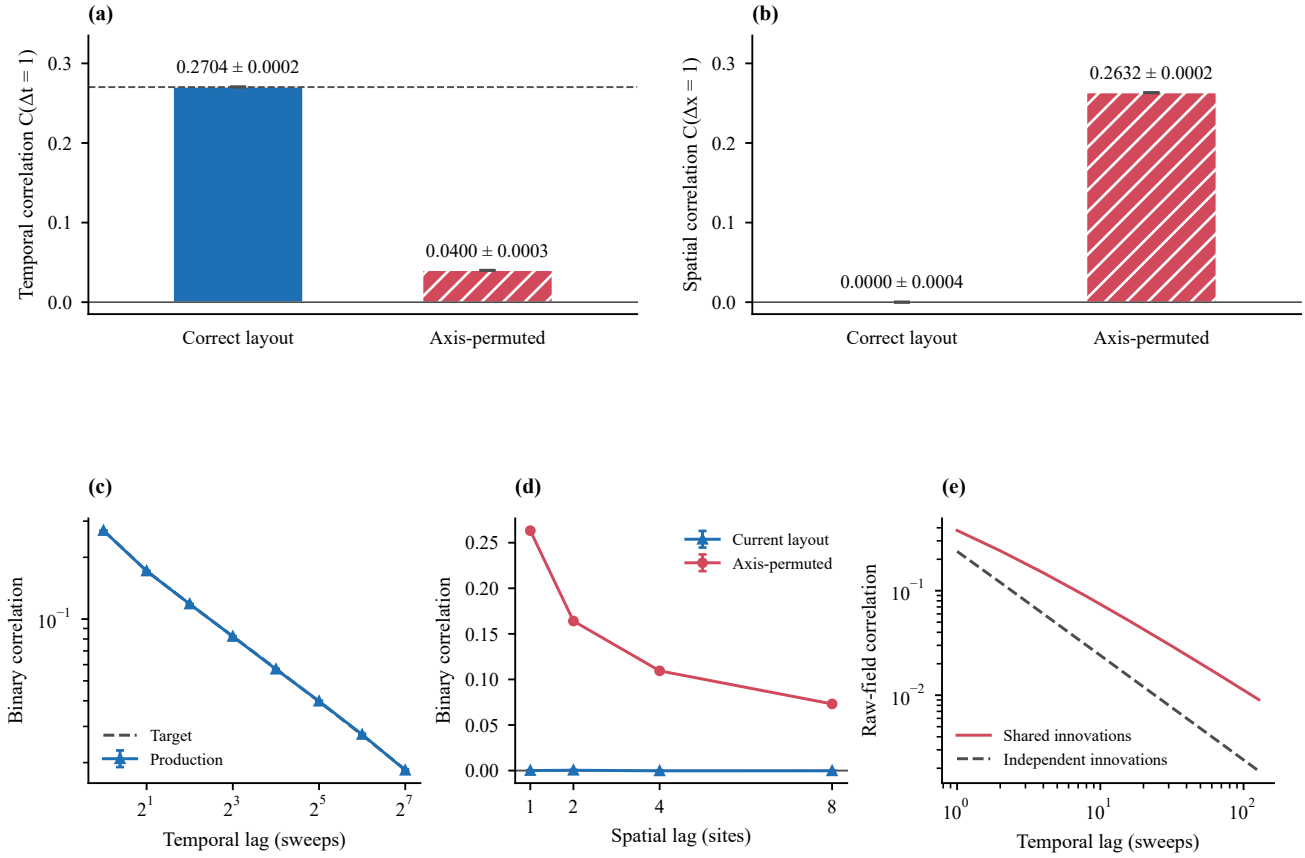


FIG. 1. Input validation from the assembled field to covariance checks. (a,b) Lag-one temporal and spatial binary correlations, respectively, under correct assignment (blue) and the axis-permuted negative control (hatched red) at $\theta = 0.25$, $L = 32$, $T = 65536$. Each pair uses the same eight runs; bar heights are run means and error bars are one run standard error. Numerical labels report mean ± standard error. The dashed line in (a) is the finite-period temporal target. The comparison shows temporal correlation weakening and spurious spatial correlation appearing under axis permutation. (c) Production binary temporal correlations for the fGn-spectrum filter at $\theta = 0.25$, $L = 64$, $T = 65536$ against its finite-period target. Points are eight-run means with one standard error. (d) Spatial correlations of the same samples at $L = 32$ under correct assignment and the axis-permuted negative control. This is the present implementation's fault control, not an attribution to historical codes. (e) Exact stationary normalised raw covariances of shared- and independently driven AR sums at $\theta = 0$, $N = 48$. No Gaussian binary conversion is applied; these are not the simulator's i.i.d. zero-parameter branch.

instead of treating a self-comparison as an independent signal experiment.

We do not infer a physical onset from the first significance crossing. A shrinking null floor can shift a crossing even when local probabilities are size independent. Conversely, local probabilities can remain fixed while spatial arrangements reorganise. Appendix C supplies a direct counterexample. No physical conclusion here relies on a novelty detector or nominal distance z scores.

IV. RESULTS

A. Spatial and temporal input checks

All eight selected spectral validation cells satisfy the stated checks at the reported lags. At $\theta = 0.25$, $L = 32$, the fGn-spectrum field has binary lag-one temporal correlation 0.2704 against finite-period target 0.2702, with spatial lag-one correlation consistent with zero. The same samples under axis permutation instead give temporal correlation 0.0400 and spatial correlation 0.2632. Figure 1(a–d) illustrates why testing the assembled field is necessary even after validating the one-dimensional filter.

Innovation covariance also matters. At nominal $\theta = 0$,

TABLE I. Five-size late-time width slopes. Every cell has 64 independent runs; errors are one run-bootstrap standard deviation from 500 resamples. Shared AR includes its i.i.d. shortcut at $\theta = 0$.

θ	Shared AR	Power law	fGn spectrum
0.00	0.503 ± 0.005	0.506 ± 0.005	0.506 ± 0.005
0.10	0.606 ± 0.006	0.542 ± 0.007	0.545 ± 0.007
1/6	0.641 ± 0.008	0.584 ± 0.006	0.588 ± 0.006
0.20	0.657 ± 0.008	0.618 ± 0.007	0.621 ± 0.007
0.25	0.717 ± 0.008	0.682 ± 0.007	0.678 ± 0.007
0.30	0.761 ± 0.008	0.740 ± 0.007	0.745 ± 0.007
0.40	0.845 ± 0.010	0.836 ± 0.009	0.833 ± 0.009

the stationary $N = 48$ AR sums have $C(1)/C(0) = 0.377315$ with shared innovations and 0.237958 with independent innovations. For the shared case, $\tau C(\tau)/C(0)$ rises from 1.04687 at $\tau = 64$ to 1.72174 at 4096; a constant-amplitude $1/\tau$ tail does not describe that window. These raw-field observations do not assign an RG phase to the binary growth model.

B. Increasing finite-size width slopes

The white-noise references give $\alpha_{\rm fit} = 0.503 \pm 0.005$, 0.506 ± 0.005 and 0.506 ± 0.005 for shared AR, power law and the fGn-spectrum input, close to 1/2 over this window. The width slopes rise with θ in all three campaigns (Fig. 2, Table I). Spatially independent spectral inputs do not produce an approximately flat 1/2 branch over this finite-size range.

At $\theta = 0.25$, shared AR and the fGn-spectrum filter give 0.717 ± 0.008 and 0.678 ± 0.007; at 0.40 they give 0.845 ± 0.010 and 0.833 ± 0.009. Thus construction changes finite-size amplitudes and slopes, but the large rise with temporal correlation also occurs under both Gaussian inputs. The maximum difference between the two spectral-filter slopes is 0.0047. Their base streams are matched, and their agreement is not independent confirmation.

Adjacent-size estimates fluctuate rather than show a monotone flow. At $\theta = 0.25$ the fGn-spectrum sequence is approximately 0.681, 0.645, 0.743, 0.626, with standard deviations about 0.030–0.035. This does not establish convergence to 1/2 or a long-range fixed point. Across all cells, the final-to-preceding-window RMS width ratio lies between 0.935 and 1.053. This checks late-time stability without proving that every finite-time correction has vanished; duration, cutoff and initialisation remain part of the protocol.

C. Paired filter comparison and duration controls

For the Gaussian filters, corresponding run records are paired by seed at each size and parameter. We resample the 64 pairs jointly within each cell, independently between sizes, and refit the difference $\delta\alpha = \alpha_{\rm fGn} - \alpha_{\rm power}$ in each of 2000 bootstrap replicates. Pointwise 95% percentile intervals contain zero at all seven parameters (Table II). At $\theta = 0.25$, $\delta\alpha = -0.00443$ with interval $[-0.01104, 0.00219]$. Thus the present ensembles do not resolve a difference between these width slopes; this is neither an equivalence test with a prespecified margin nor independent replication of a scaling law.

TABLE II. Paired Gaussian-filter slope differences and pointwise 95% run-bootstrap percentile intervals. Sizes and central estimators are the same as in Table I.

θ	$\delta\alpha$	95% interval
0.00	−0.00010	$[-0.00240, 0.00220]$
0.10	0.00266	$[-0.00366, 0.00932]$
1/6	0.00390	$[-0.00289, 0.01081]$
0.20	0.00269	$[-0.00510, 0.01010]$
0.25	−0.00443	$[-0.01104, 0.00219]$
0.30	0.00468	$[-0.00100, 0.01037]$
0.40	−0.00333	$[-0.00869, 0.00184]$

To separate observation duration from changes in spectral cutoff, we also generate one length-$4T$ fGn-spectrum record per run at each combination of $\theta = 0.10, 0.40$ and $L = 128, 256$. Here T is the original spectral campaign duration for that size. Each of 32 new independent runs starts from a flat interface and uses nested prefixes of the same record to reach T, $2T$ and $4T$. The terminal fifth of regularly sampled squared widths defines each estimate. Width ratios use 2000 paired run-bootstrap resamples. The cutoff remains $1/(4T)$ throughout a cell, so its T prefix is a separate control, not a repeat of the original length-T periodic input. Figure 3 reports the resulting ratios and uncertainties. At $4T$, the four central ratios range from 0.989 to 1.034. For $L = 256$, they are 1.031 ± 0.025 at $\theta = 0.10$ and 1.034 ± 0.033 at 0.40, where errors are one bootstrap standard deviation. All eight pointwise 95% ratio intervals contain unity. No duration-induced width change is resolved at this precision in these controls; this does not demonstrate saturation or exclude smaller drifts. These controls test observation-time sensitivity under a fixed input record; they do not establish convergence with respect to the spectral cutoff.

D. Local morphology and detection floors

At $L = 256$, the flat-pair fraction decreases as correlations strengthen for every input [Fig. 4(a)]. For the fGn-spectrum filter it changes from 0.4364 ± 0.0044 at $\theta = 0$ to 0.1481 ± 0.0044 at 0.40; shared AR changes from 0.4233 ± 0.0042 to 0.1138 ± 0.0035. There is no opposite-sign response between the correctly assembled inputs on this grid. At $\theta = 0.40$, $L = 256$, the shared-AR and fGn-spectrum late-time widths are 17.97 and 13.80, a ratio of 1.30. Figure 5 shows their interface evolution as continuous surfaces; Fig. 4(c) retains the ensemble spa-

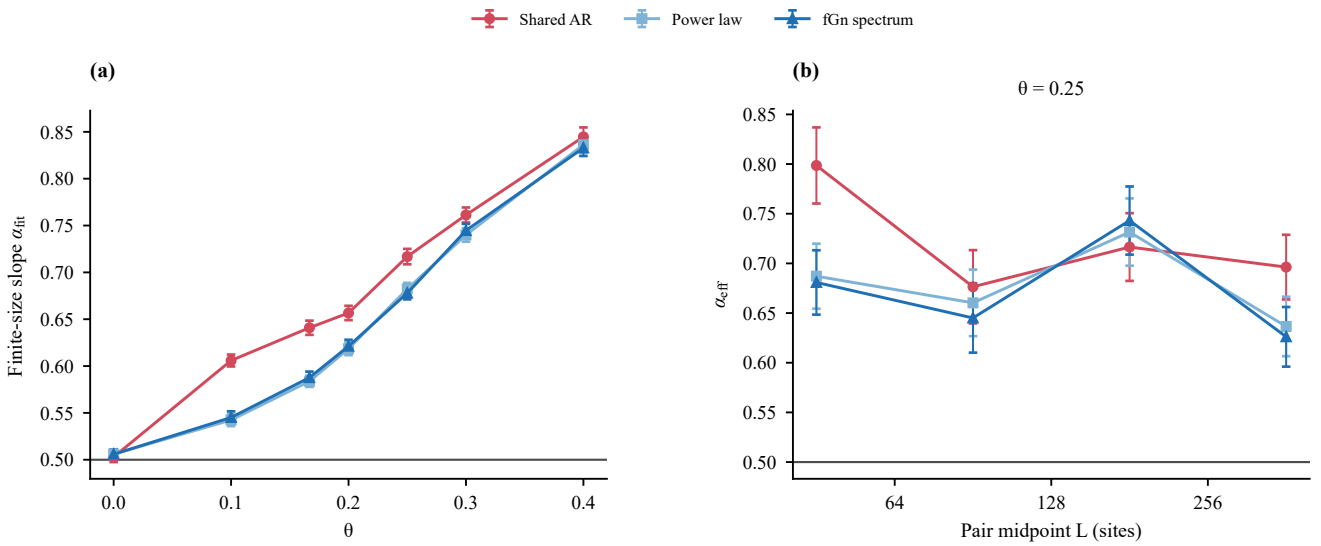


FIG. 2. Regenerated width slopes. (a) $\alpha_{\rm fit}$ with one run-bootstrap standard deviation. The horizontal 1/2 line is a reference, not a fitted phase boundary. (b) Adjacent-size $\alpha_{\rm eff}$ at $\theta = 0.25$, with bootstrap errors, at geometric size-pair midpoints. All four pairs are shown. These finite-window quantities do not measure asymptotic exponents or θ_c.

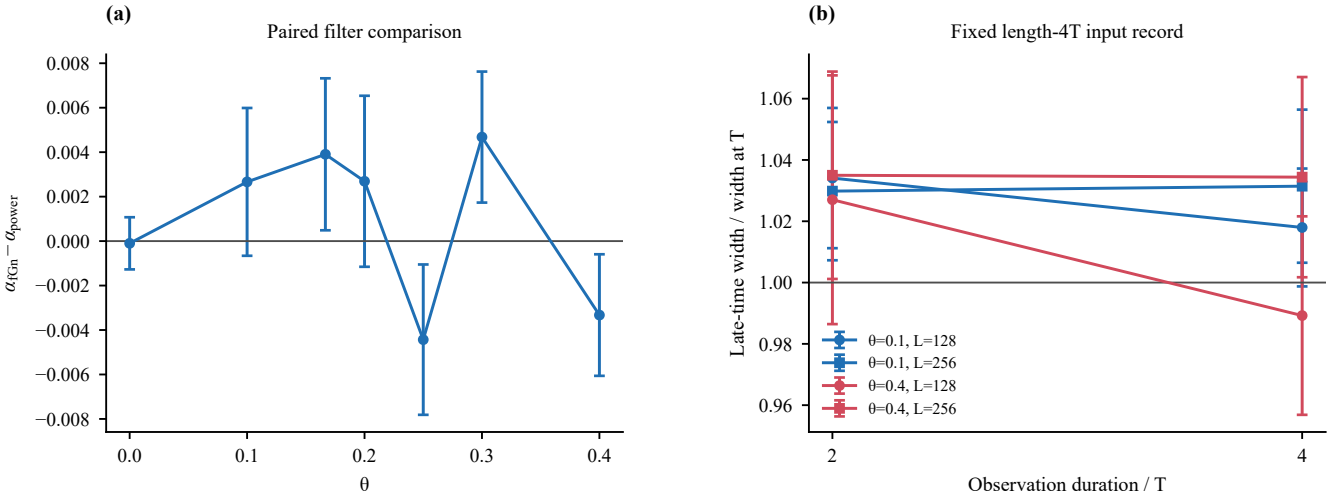


FIG. 3. Additional inference and duration controls. (a) Paired Gaussian-filter slope differences with one bootstrap standard deviation; Table II reports percentile intervals. (b) Late-time RMS width at $2T$ and $4T$, divided by its value at T, using nested prefixes of one fixed length-$4T$ fGn-spectrum record per run. Colours distinguish θ and markers distinguish sizes. Errors are one paired run-bootstrap standard deviation from 32 independent runs per cell.

tial spectra. Figure 6 complements this comparison with surface views at weak, intermediate and strong temporal correlation. These surfaces visualise the arrangement and persistence of height fluctuations; they are illustrative single runs, not estimates of ensemble differences. A single size and spectrum are not used to identify a faceting or anomalous-scaling boundary.

The Hellinger excess detects local changes under correlated driving [Fig. 4(b)], but its magnitude and size dependence do not define a unique physical onset. In the fGn-spectrum reference, the conditional null mean changes from 0.0155 at $L = 32$ to 0.0036 at 512, a factor of 4.31, despite matched run counts. Intermediate sizes need not be perfectly monotone. The number of

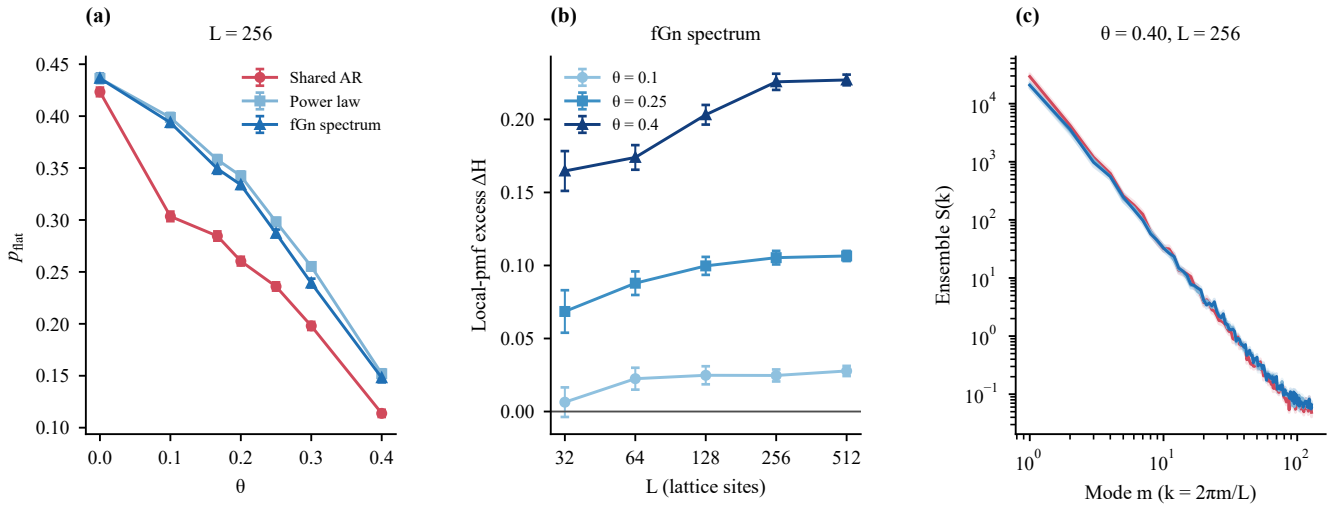


FIG. 4. Local morphology and pmf distances. (a) Flat-pair fraction at $L = 256$ from all 64 final profiles per cell, with run-bootstrap standard deviations. Every input shows a decrease on this grid. (b) Excess Hellinger distance in the fGn-spectrum campaign at three fixed θ values, with outer independent-run bootstrap standard deviations from 200 resamples. Conditional partition scatter is not substituted for these errors. This local statistic is not a necessary or sufficient test for a universality crossover. (c) Mean $S(k) = L^{-1}|\hat{h}(k)|^2$ at $\theta = 0.40$, $L = 256$ over all 64 final profiles, with one run standard error shaded. Colours identify the same inputs as in (a). All positive Fourier modes are shown without smoothing or an artificial floor; no asymptotic spectral exponent is fitted.

columns and their spatial correlations affect each run's pooled-pmf precision.

A first crossing of a fixed number of null standard deviations mixes local probability changes with detection power. This statement concerns the estimator used here. Without reconstructing another study's observable, samples and decision rule, it does not establish that a reported threshold has the same origin. A detectable local difference likewise does not demonstrate a change of scaling universality.

E. Stationary innovation and covariance-matching controls

The additional grid uses all five sizes $L = 32, 64, 128, 256, 512$. At $\theta = 0.10, 0.25, 0.40$, we compare shared and independent AR innovations under the same weights and add covariance-compensated independent modes. Both centred uniform and Gaussian innovations are used, giving six inputs with 64 runs per cell. Every input is a circular length-$2T$ record with its DC component removed. Here T is the next 5-smooth integer at or above $15L^{3/2}$, and the mode count is determined by $2T$. Flat interfaces evolve under one record to T and $2T$. Late-time squared widths are averaged over the last 20% of regularly stored times before taking the ensemble square root. This campaign controls initialization and cutoff within its own comparisons; it does not replace the cold-start protocol of the original shared-AR campaign.

For the positive weights in Eq. (7), define

$$A_n = W_n \sum_m \frac{W_m}{1 - r_n r_m}, \qquad V_n = \sqrt{(1 - r_n^2) A_n}. \tag{20}$$

Independent innovations with weights V_n reproduce the shared-input stationary covariance at every lag, not just its variance. The spectral identity survives sampling on a common frequency grid and removing DC. Zero-mean Gaussian processes with the same complete covariance have the same finite-dimensional laws; their thresholded fields and resulting BD observables therefore also agree in law. The Gaussian comparison is an implementation and sampling control, rather than evidence for an additional physical effect of a latent mode representation.

Figure 7(a) and Table III quantify how innovation coupling and covariance compensation affect the effective width slopes. Raw-covariance compensation cannot by itself match the actual uniform-driven binary input: the maximum measured sign-correlation residual across the control cells is 0.039–0.052. Thresholding a non-Gaussian process depends on its two-time orthant probabilities, which are not fixed by its raw covariance.

We therefore add a Gaussian surrogate fitted to the *binary* covariance. Only the first 32 uniform-shared noise records in each cell calibrate a nonnegative mixture of circular AR spectra, using the Gaussian relation $\rho(\tau) = \sin[\pi C_s(\tau)/2]$. Fresh Gaussian and uniform validation records, with 64 runs and 16 sites per run, test

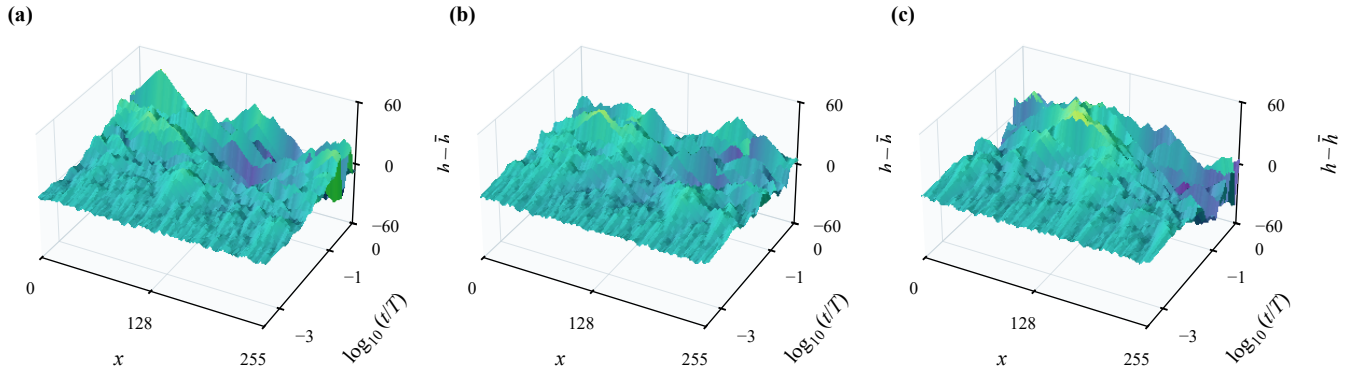


FIG. 5. Continuous surface views of interface evolution at $\theta = 0.40$, $L = 256$. Panels (a–c), from left to right: shared AR, power law and the fGn-spectrum input. Each surface spans position $x = 0, \ldots, 255$ horizontally, time $\log_{10}(t/T)$ in depth (earlier profiles in front, later profiles at the back), and height $h(x,t) - \bar{h}(t)$ vertically. These are one-dimensional interfaces evolving in time. Each representative run has final width nearest the 64-run median. All 32 archived times and 256 sites are retained as surface vertices; faces connect neighbouring samples linearly and do not supply additional simulated times. Height and colour limits are common to all panels, $[-60, 60]$ lattice units. Purple/blue denote lower heights and green/yellow higher heights; lighting accentuates surface orientation. No amplitude rescaling is applied.

every integer lag from 0 through 64 and the sampled long lags. A simultaneous bootstrap upper bound on $\max_\tau |\Delta C_s(\tau)|$ must fall below 0.005, an explicit numerical tolerance rather than a universal physical equivalence criterion. Two initial cells, $\theta = 0.40$ at $L = 32, 64$, failed this test. We performed one documented calibration refinement, retaining the same spectral family and tolerance: 128 fresh 64-site records calibrate every integer lag from 0 through 64 and the original long lags, followed by 64 new, independent 64-site validation records per cell. The initial failures motivated this refinement. After its calibration scheme was fixed, the fresh validation records were not used for fitting, parameter selection or further retries; the tolerance was not relaxed. The simultaneous upper bounds changed as follows: $L = 32$: $0.01074 \to 0.00371$; $L = 64$: $0.00585 \to 0.00255$. At $\theta = 0.40$, $L = 512$, the initial upper bound was 0.00531. Keeping the fitted spectrum and all interface records fixed, we performed one separate covariance-only validation with 128 fresh runs and 64 sites per input and run. Its upper bound is 0.00250. These records were not used to fit the spectrum or interface slopes. No further validation retries or tolerance relaxation were used. With these refined controls, 15 of 15 reported cells pass. The original failed archives remain in the numerical supplement. Every reported binary-control cell passes the retained validation tolerance.

The surrogate interface comparison uses the 32 held-out uniform-shared runs. Among parameter grids passing every covariance check, the surrogate slope contrasts at $2T$ are unresolved within the pointwise intervals in Table III. An unresolved difference does not prove equal exponents. Independent four-time input statistics can nevertheless differ. Sign-flip symmetry and stationarity imply

$$\langle s_0 s_1 s_2 s_3 \rangle = 16P(++++) - 1 - 3C_s(1) - 2C_s(2) - C_s(3). \tag{21}$$

Figure 7(b) evaluates this moment with fresh validation records at $L = 256$, where all three covariance checks pass. The $\theta = 0.25$ and 0.40 four-time input differences are resolved. These observations distinguish a detectable high-order input change from a demonstrated change in interface scaling. The original failed controls are excluded from mechanistic inference. Only parameter grids passing every retained covariance check are eligible for that comparison. Unmeasured lag residuals and uncertainty in the fitted spectrum still preclude a stronger attribution. Intervals use 2000 whole-run bootstrap draws, preserve the six-input pairing within size, and resample sizes independently. Surrogate intervals are conditional on the fitted spectrum; no multiple-comparison adjustment is applied to these exploratory contrasts. The raw run archives, original failures and reproducible analysis and figure code are provided in the Supplemental Material [13].

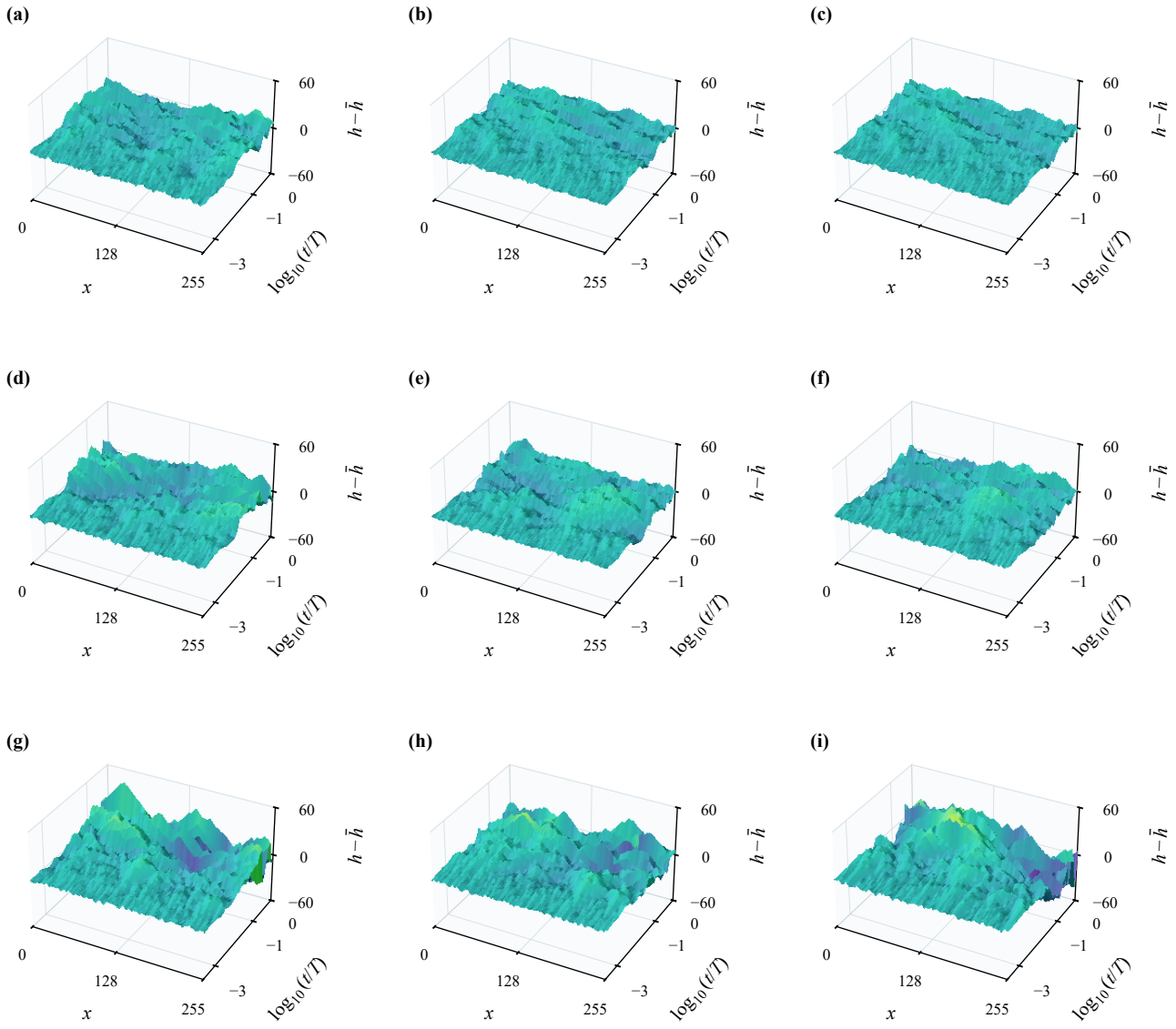


FIG. 6. Continuous surface morphology at $L = 256$. Panels (a–i) are ordered row-wise. Columns, from left to right, distinguish shared AR, power law and the fGn-spectrum input; rows, from top to bottom, show $\theta = 0.10$, 0.25 and 0.40. Each cell displays all 256 sites and all 32 stored profiles of the run with final width nearest its own 64-run median. At each time the spatial mean height is subtracted. Position, logarithmic time and centred height have the same orientation as in Fig. 5. A common height range and colour scale of $[-60, 60]$ lattice units is used across all nine panels without amplitude rescaling. Purple/blue denote lower heights and green/yellow higher heights. Time is normalised by each cell's archived duration T. Surface faces connect neighbouring stored samples linearly; no additional dynamical data or spatial smoothing is introduced. These representative runs illustrate morphology, not an ensemble-averaged effect or a scaling boundary.

V. DISCUSSION

The central result is a finite-window comparison in which input validation and statistical comparison are treated separately. Both Gaussian spectral filters yield increasing width slopes with temporal memory, and paired resampling does not resolve their slope differences at the tested parameters. The shared AR variant differs most visibly at weak and intermediate memory. Agreement between the Gaussian filters therefore does not imply insensitivity to every noise construction, while their common trend shows that increasing width slopes are not exclusive to shared AR innovations.

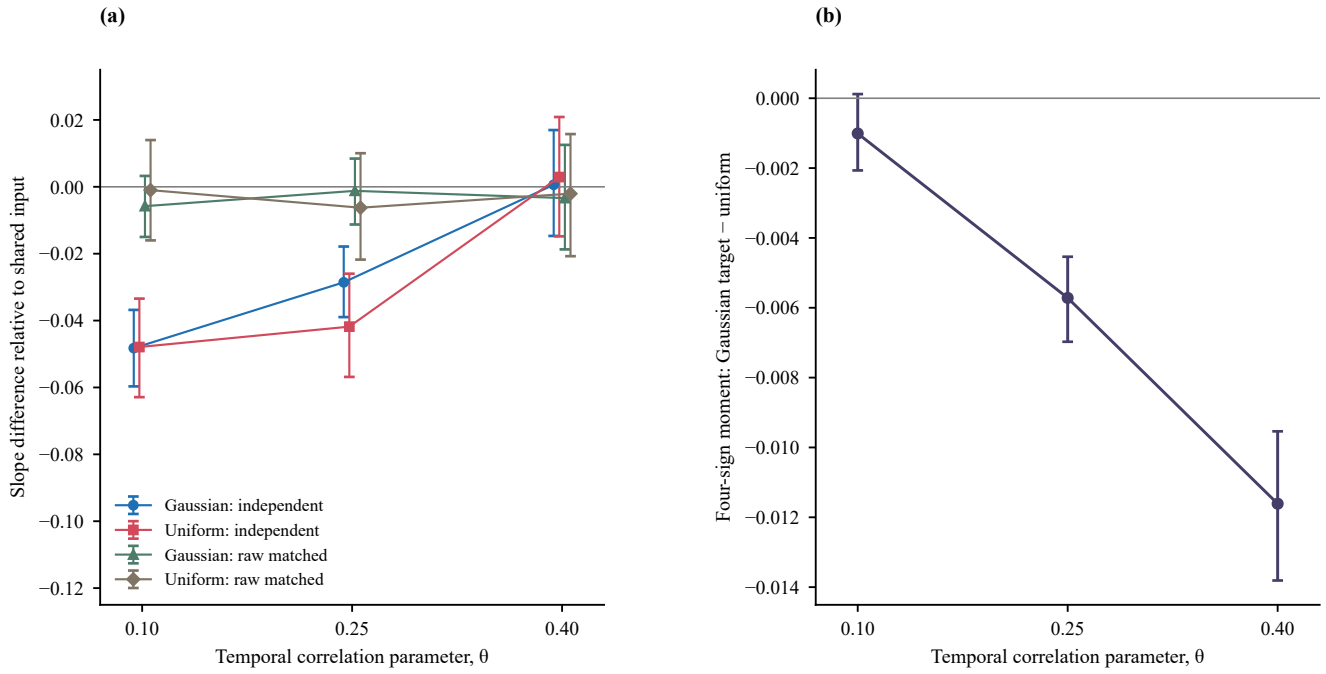


FIG. 7. Stationary innovation controls. (a) Finite-size slope differences at $2T$ relative to the corresponding shared-input family. Each estimate uses 5 sizes ($L = 32$–512) and 64 runs per input and size. (b) Gaussian binary-target minus fresh uniform-shared consecutive-four-sign moment at $L = 256$, reconstructed by Eq. (21); 64 independent validation runs per input, 16 sites per run. Bars are pointwise 95% whole-run bootstrap intervals from 2000 draws. High-order input differences and finite-size slope sensitivity are distinct observables; approximate covariance matching does not establish identical full input processes or an asymptotic universality class.

TABLE III. Innovation-control slope differences at duration $2T$ across $L = 32$–512. G and U denote Gaussian and uniform innovations. Independent inputs are compared with their corresponding shared input; the binary-target Gaussian surrogate is compared with the held-out uniform-shared runs. Intervals are pointwise 95% run-bootstrap intervals.

θ	Input contrast	$\Delta\alpha_{\mathrm{eff}}$	95% interval
0.10	G independent	−0.0482	$[-0.0597, -0.0368]$
0.10	U independent	−0.0479	$[-0.0629, -0.0334]$
0.10	G binary target	−0.0068	$[-0.0244, 0.0094]$
0.25	G independent	−0.0286	$[-0.0390, -0.0179]$
0.25	U independent	−0.0418	$[-0.0569, -0.0260]$
0.25	G binary target	−0.0175	$[-0.0365, 0.0011]$
0.40	G independent	0.0006	$[-0.0147, 0.0170]$
0.40	U independent	0.0029	$[-0.0148, 0.0209]$
0.40	G binary target	−0.0133	$[-0.0340, 0.0084]$

The validation controls identify two distinct requirements. First, correlation measurements must reach the final space–time field passed to the growth rule. An axis-assignment fault can preserve filtered sequences while altering the coordinates in which their correlations act. Second, an AR specification must include the covariance of its innovations across modes. Identical mode weights with shared and independent innovations describe different input processes. The stationary campaign in Sec. IV E separates this change of process from a covariance-compensated representation. For Gaussian inputs, complete covariance compensation restores the full process law. Uniform innovations require the additional check of the thresholded covariance; matching the latent field alone leaves measurable binary-input differences.

The fixed-record controls address observation duration without changing the lowest available input frequency within each cell. Their interpretation is correspondingly restricted to that input record and to the tested sizes and parameters. They do not control the difference between periodic spectral inputs and cold-started AR streams. Residual differences can also involve short-time correlations and non-Gaussian marginals. Matching nominal θ is insufficient to isolate any of these mechanisms. The additional stationary controls remove this initialization mismatch within their own grid, while retaining explicit finite-lag validation limits. Their four-time input statistics illustrate why a measurable high-order difference is not itself evidence for a changed roughness exponent.

Local distances answer a different question from spa-

tial scaling. Equation (17) and its expansion show that the increment-pmf Hellinger distance measures changes in one local probability. Its sampling floor affects detection power, but the distance discards the arrangement of steps. The explicit counterexample in Appendix C shows why neither a constant distance nor a detectable departure identifies a scaling universality class. Comparisons with emergence diagnostics [7] must retain the observable and sampling rule of each study.

Microscopic sensitivity under temporal correlations has already been emphasised [8]. The present work adds an explicit field-assembly control, innovation-covariance specification, paired filter inference and fixed-record duration checks in one discrete model. The production grid stops at $L = 512$ and uses durations proportional to $L^{3/2}$; bootstrap errors do not quantify residual cutoff or initialization effects. Establishing asymptotic exponents or a faceting boundary would require larger sizes, independently controlled cutoffs and stationary input comparisons. The different threshold definitions in the literature therefore remain separate physical questions, and no historical threshold is invalidated here.

VI. CONCLUSION

Innovation coupling and high-order noise statistics affect different levels of the driving process. We construct independent AR weights that compensate the complete shared-input covariance; for Gaussian innovations this fixes the entire process law and hence the distribution of BD observables. Uniform innovations show why latent covariance compensation is insufficient after binary thresholding. The additional binary-target controls validate 15 of 15 cells at the unchanged 0.005 tolerance, with the original failures and their independent recalibration retained for audit. At $\theta = 0.25$, $L = 256$, a four-time input difference remains resolved under this validation, while the corresponding effective-slope contrast across $L = 32$–512 is not resolved. This distinguishes measurable high-order temporal structure from evidence for changed roughness scaling; an unresolved contrast is not proof of physical equivalence.

The original three-input comparison, assembled-field checks, paired filter inference and fixed-record duration controls support this interpretation. The conclusions remain conditional on the fitted surrogate spectrum, the sampled covariance lags and finite observation windows. They do not establish asymptotic exponents, a universality-class change or a critical temporal-memory parameter.

DATA AVAILABILITY

The code, production data and additional control results are maintained in `https://github.com/HuangYong2006/bd-correlated-noise-1.0.0-code`. The reproducibility materials contain all 105 production cells, validation outputs, analysis and plotting scripts, seeds, layout identifiers and source hashes. The reproduction guide maps figures and tables to their inputs. The stationary controls added in Sec. IV E form a separate revision data set, packaged as `innovation_controls_data_20261031.zip`; they are not included in the earlier production release. This numerical supplement accompanies the manuscript and contains the run-level archives, original failed controls, refined controls, calibration metadata, seeds, analysis code, a standalone numerical verifier and a file-level SHA256 manifest [13].

Appendix A: Single-step constraint

If all neighbour differences have magnitude at most one, Eq. (3) gives $h(x,t) \le h(x,t+1) \le h(x,t)+1$. The maximum at the next site includes $h(x,t)$, so $h(x+1,t+1) \ge h(x,t)$ and $h(x,t+1) - h(x+1,t+1) \le 1$. Exchanging sites gives the reverse bound. Induction from the flat state proves the constraint. Periodic summation forces equal up- and down-step counts. No height-inversion symmetry of the maximum update is needed.

Appendix B: Covariance and limiting behaviour

Stationarity gives $\Sigma_{nm} = r_n r_m \Sigma_{nm} + Q_{nm}$, hence $\Sigma_{nm} = Q_{nm}/(1 - r_n r_m)$ and Eq. (8). With zero initial modes and t completed innovations, the covariance instead is

$$C_N(t,\tau) = \sum_{n,m} W_n W_m r_n^{\tau} Q_{nm} \frac{1-(r_n r_m)^t}{1 - r_n r_m}. \tag{B1}$$

A finite cold-start record is not silently identified with a stationary one. The stationary shared-innovation spectrum is

$$S_N^{\mathrm{shared}}(\omega) = \sigma_z^2 \left| \sum_n \frac{W_n}{1 - r_n e^{-i\omega}} \right|^2, \tag{B2}$$

whereas independent innovations give a sum of squared transfer functions. Cross terms cannot be discarded in the shared case.

The impulse response $g_j = \sum_n W_n r_n^j$ has continuum-grid approximation $g_j \propto j^{\theta-1}$ for intermediate $u_N j \ll 1 \ll u_1 j$. At $\theta = 0$, a $1/j$ kernel combined with

$$\sum_{j=1}^{\infty} \frac{1}{j(j+\tau)} = \frac{H_\tau}{\tau}, \qquad H_\tau \simeq \ln \tau + \gamma, \tag{B3}$$

permits a logarithmic correction. This mechanism argument is not an exact finite-N pure-power asymptote: every finite-mode covariance ultimately decays exponentially. Convergence at fixed lags does not exchange infinite-mode and infinite-lag limits.

Appendix C: Information missing from a local pmf

On an even periodic lattice, consider equal counts of $+1$ and -1 increments without zeros. Alternating signs and a block of $L/2$ positive followed by $L/2$ negative increments have identical nearest-neighbour pmfs. Yet the alternating interface has bounded width and the block arrangement has width proportional to L. These are information-theoretic examples, not assertions about stationary states of Eq. (3). They show why the single-step constraint does not make a local pmf sufficient for large-scale roughness. Constant ΔH cannot rule out a crossover, and increasing ΔH cannot alone establish one.

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[13] See Supplemental Material at [URL will be inserted by publisher] for the run-level innovation-control archives, initial covariance-validation failures, refined and independent validation records, calibration metadata, seeds, reproducible analysis and figure code, and file-level checksums.